CS670 Mid-Term Project Report

Privacy-Preserving Recommendation System: OblivRec

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1. Project Status

The literature survey, *Privacy-Preserving Recommendation Systems: A Survey of Cryptographic Approaches*, is complete, along with its accompanying presentation video (https://youtu.be/5_uQ8ESewRk). Building on this, we developed **OblivRec**, a working demonstrator that combines the three-party private-training design of *Nudge* with the private-delivery design of *Pirsona*. Since neither paper specifies how the two compose end-to-end, integrating them and validating the result experimentally is the project's main contribution at this stage.

This stage covers private serving and delivery only; the model itself is still trained in the clear, with private training planned for the next phase. The demonstrator scores each user against the MovieLens-100K catalogue using 2-of-3 replicated secret sharing, and retrieves recommended items through a self-built distributed-point-function (DPF) private-information-retrieval (PIR) layer, ensuring that no server learns the user's data or which items were retrieved. A 20-page technical companion report provides full derivations, correctness proofs, and benchmark data.

Two results are notable. First, private delivery is essential: because the item-embedding matrix is public, an observer who sees which items a user fetches can reconstruct their preferences with high accuracy, predicting 56% of a user's next twenty recommendations after only ten fetches; adding decoy traffic fails to close this gap even at five times the real volume. Second, private retrieval sends 446× fewer bytes than downloading the full catalogue; weighed against its higher server-side computation, it remains the better overall choice on any link slower than roughly 8 Gbit/s, which is to say on any real network. A trained statistical distinguisher confirmed that neither server can identify which item was fetched from network traffic alone.

2. Individual Contributions

Mainak Sarkar (230619) – DPF construction, correctness, and hardening

- Implemented a distributed point function (DPF) from scratch, sharing one traversal implementation across single-point and whole-domain evaluation.
- Built an exhaustive correctness test suite (7.16 billion leaf comparisons, covering every input up to 16 domain bits on the 64-bit ring) and fixed a key-deserialisation bug that silently accepted an invalid party byte.
- Replaced an insecure random-number generator with an OS-level CSPRNG and hardened the build with undefined-behaviour trapping, checked STL containers, and a bounds-checked span type.
- Verified via a statistical distinguisher that an eavesdropper cannot identify a fetched item from raw network traffic.

Shravan Agrawal (230984) – Private serving and delivery pipeline

- Implemented the MovieLens-100K record packing and the DPF-based PIR layer for private item retrieval.
- Implemented the 2-of-3 secret-shared scoring engine, including seen-item masking, using the public item-embedding matrix to keep scoring communication-free.
- Built and validated the float-to-ring encoding, fixing a rounding-mode and a quantisation-order bug, and verified the demo matches the cleartext baseline bit-for-bit.
- Implemented the length-prefixed TCP framing layer connecting the three servers and the client, with validation against partial reads/writes and untrusted inputs.

Shrasti Dwivedi (230982) – Modeling, benchmarking infrastructure, and coordination

- Implemented a cleartext power-iteration baseline and showed recommendation quality remains stable even before full convergence, justifying a cheaper training configuration.
- Built the reproducible benchmarking harness used to measure DPF performance.
- Prepared the literature-survey presentation, coordinated its submission, and led the writing of this report.
- Designed the reconstruction-attack experiment across all 943 users (with popularity and random-direction controls) and showed that a decoy-based defence fails to close the privacy gap even at five times the real traffic.